LoRA Fine-Tuning of GPT-2 on Multi-Domain Text Datasets

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Abstract

This report presents parameter-efficient fine-tuning of GPT-2 using Low-Rank Adaptation (LoRA) across a curated collection of multi-domain, text-only datasets. Two LoRA configurations (r=8 and r=4) were evaluated to study the efficiency–performance trade-off. The lower rank configuration achieved improved generalization while reducing trainable parameters.

1 Methodology

GPT-2 small (124M parameters) was fine-tuned using the Parameter-Efficient Fine-Tuning (PEFT) framework with LoRA. Instead of updating all model weights, only low-rank adaptation matrices were trained within attention layers (specifically the `c_attn` module).

Two configurations were evaluated:

- LoRA Rank r = 8
- LoRA Rank r = 4

Reducing rank decreases trainable parameters while maintaining adaptation capability.

2 Datasets

A curated mixture of 10 text-only datasets was used to ensure diversity across encyclopedic, narrative, conversational, summarization, and review-based domains. All datasets were sampled to remain compatible with T4 GPU memory constraints.

The following datasets were used:

- **WikiText-2 (wikitext-2-raw-v1)** – Wikipedia-derived language modeling dataset
- **WikiText-103 (wikitext-103-raw-v1)** – Large-scale Wikipedia corpus

- **TinyStories** – Synthetic short narrative stories
- **AG News** – News topic classification corpus
- **XSum** – Extreme summarization dataset (document field used)
- **CNN/DailyMail (3.0.0)** – News articles (article field used)
- **SQuAD** – Question-answering dataset (context field used)
- **Yelp Review Full** – Long-form customer reviews
- **IMDB** – Movie reviews dataset
- **Menlo Instruction Text Only** – Instruction-style text corpus

Each dataset was preprocessed to extract raw text fields only and tokenized to a maximum sequence length of 256 tokens.

3 Training Setup

- Base Model: GPT-2 (124M parameters)
- Epochs: 2
- Batch size: 4
- Learning rate: 5×10^{-5}
- Weight decay: 0.01
- Evaluation strategy: Per epoch
- Hardware: NVIDIA T4 GPU (Google Colab)
- Experiment tracking: Weights & Biases

4 Results

- LoRA r=8: Perplexity ≈ 76
- LoRA r=4: Perplexity = 18.94

The reduction in LoRA rank resulted in approximately 50% fewer trainable parameters while significantly improving validation perplexity, indicating improved generalization on the multi-domain dataset mixture.

5 Inference Performance

Inference benchmarking on T4 GPU showed an average throughput of approximately 50 tokens per second for generation with 100 new tokens.

6 Discussion

The results demonstrate that lower-rank LoRA adaptation can improve generalization in small language models when trained on diverse multi-domain corpora. The r=4 configuration reduced parameter count while improving perplexity, suggesting reduced overfitting and better parameter efficiency.

7 Conclusion

This study highlights the effectiveness of parameter-efficient fine-tuning using LoRA for small language models under constrained hardware settings. The experiments confirm that careful rank selection significantly impacts both computational efficiency and model generalization performance.